Factors, Multiples, Squares and Cubes

Name _____ Date _____ No calculator Version 1 Scan to practise this topic online

KEY WORDS factor *(tauwehe)* · factor pair · multiple *(tau whakarea)* · square number *(tau pūrua)*

A Factors and factor pairs ●○○

Remember: A factor of a number divides into it exactly. Factors come in pairs: $12 = 1 \times 12 = 2 \times 6 = 3 \times 4$.

1. Write all the factor pairs of each number.

a) 12 _____ **b)** 18 _____

c) 20 _____ **d)** 30 _____

2. List all the factors of each number.

a) 8 _____ **b)** 15 _____

c) 24 _____ **d)** 36 _____

e) 13 _____ **f)** 40 _____

B Multiples ●○○

Watch out: Factors are the numbers that divide into a number. Multiples are the numbers in its times table. The factors of 6 are 1, 2, 3, 6. The multiples of 6 are 6, 12, 18, ...

3. Write the first five multiples of each number.

a) 3 _____ **b)** 7 _____

c) 9 _____ **d)** 12 _____

4. Circle the multiples in each list.

a) Multiples of 4: 6 12 18 20 30 36 42

b) Multiples of 6: 12 16 18 26 30 42 56

c) Multiples of 9: 18 24 27 39 45 54 63

C Square numbers ●●○

Remember: A square number is a number multiplied by itself: $7^2 = 7 \times 7 = 49$.

5. Work out.

a) 3^2 = _____ **b)** 6^2 = _____ **c)** 1^2 = _____ **d)** 9^2 = _____

e) 4^2 = _____ **f)** 11^2 = _____ **g)** 7^2 = _____ **h)** 12^2 = _____

i) 5^2 = _____ **j)** 10^2 = _____ **k)** 8^2 = _____ **l)** 2^2 = _____

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6. Circle the square numbers.

a) Square numbers: 4 12 25 48 64 90 100 121

D Cube numbers ●●○

⚠ **Watch out:** 2^3 means $2 \times 2 \times 2 = 8$. It does not mean 2×3.

7. Work out.

a) 1^3 = _____ **b)** 2^3 = _____ **c)** 3^3 = _____ **d)** 4^3 = _____

e) 5^3 = _____ **f)** 10^3 = _____ **g)** $2^3 + 3^2$ = _____ **h)** $5^3 - 5^2$ = _____

8. Circle the cube numbers.

a) Cube numbers: 1 9 27 36 64 100 125 150

E Squared and cubed ●●○

💡 **Remember:** "5 squared" is written 5^2 and means 5×5. "4 cubed" is written 4^3 and means $4 \times 4 \times 4$.

9. Write each one using a small number (an exponent), then work it out.

a) 5 squared **b)** 4 cubed **c)** 9 squared

d) 2 cubed **e)** 12 squared **f)** 3 cubed

F Problems ●●●

10. A square garden has sides of 8 m. How many square metres of grass are needed to cover it?

11. A cube is built from small cubes. It is 3 cubes long, 3 cubes wide and 3 cubes high. How many small cubes are used?

12. 24 chairs are set out in equal rows. Write all the ways this can be done, as rows × chairs in each row.

★ Extension

13. Which number less than 100 is both a square number and a cube number, and is not 1?

14. Find a number that is a multiple of both 4 and 6. What is the smallest one?

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Factors, Multiples, Squares and Cubes

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KEY WORDS factor *(tauwehe)* · factor pair · multiple *(tau whakarea)* · square number *(tau pūrua)*

A Factors and factor pairs ●○○

Remember: A factor of a number divides into it exactly. Factors come in pairs: $12 = 1 \times 12 = 2 \times 6 = 3 \times 4$.

1. Write all the factor pairs of each number.

a) 10 _____ **b)** 16 _____

c) 28 _____ **d)** 24 _____

2. List all the factors of each number.

a) 6 _____ **b)** 21 _____

c) 32 _____ **d)** 48 _____

e) 11 _____ **f)** 45 _____

B Multiples ●○○

Watch out: Factors are the numbers that divide into a number. Multiples are the numbers in its times table. The factors of 6 are 1, 2, 3, 6. The multiples of 6 are 6, 12, 18, ...

3. Write the first five multiples of each number.

a) 4 _____ **b)** 6 _____

c) 8 _____ **d)** 11 _____

4. Circle the multiples in each list.

a) Multiples of 3: 9 14 18 22 27 31 36

b) Multiples of 7: 14 21 27 35 44 49 54

c) Multiples of 8: 16 20 24 36 40 56 62

C Square numbers ●●○

Remember: A square number is a number multiplied by itself: $7^2 = 7 \times 7 = 49$.

5. Work out.

a) 4^2 = _____ **b)** 7^2 = _____ **c)** 1^2 = _____ **d)** 8^2 = _____

e) 3^2 = _____ **f)** 12^2 = _____ **g)** 6^2 = _____ **h)** 11^2 = _____

i) 5^2 = _____ **j)** 10^2 = _____ **k)** 9^2 = _____ **l)** 2^2 = _____

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6. Circle the square numbers.

a) Square numbers: 9 15 36 50 49 80 81 144

D Cube numbers ●●○

⚠ **Watch out:** 2^3 means $2 \times 2 \times 2 = 8$. It does not mean 2×3.

7. Work out.

a) 2^3 = _____ **b)** 1^3 = _____ **c)** 4^3 = _____ **d)** 3^3 = _____

e) 5^3 = _____ **f)** 10^3 = _____ **g)** $3^3 + 2^2$ = _____ **h)** $4^3 - 4^2$ = _____

8. Circle the cube numbers.

a) Cube numbers: 8 16 25 27 50 64 100 125

E Squared and cubed ●●○

💡 **Remember:** "5 squared" is written 5^2 and means 5×5. "4 cubed" is written 4^3 and means $4 \times 4 \times 4$.

9. Write each one using a small number (an exponent), then work it out.

a) 6 squared **b)** 5 cubed **c)** 8 squared

d) 3 cubed **e)** 11 squared **f)** 2 cubed

F Problems ●●●

10. A square garden has sides of 9 m. How many square metres of grass are needed to cover it?

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★ Extension

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